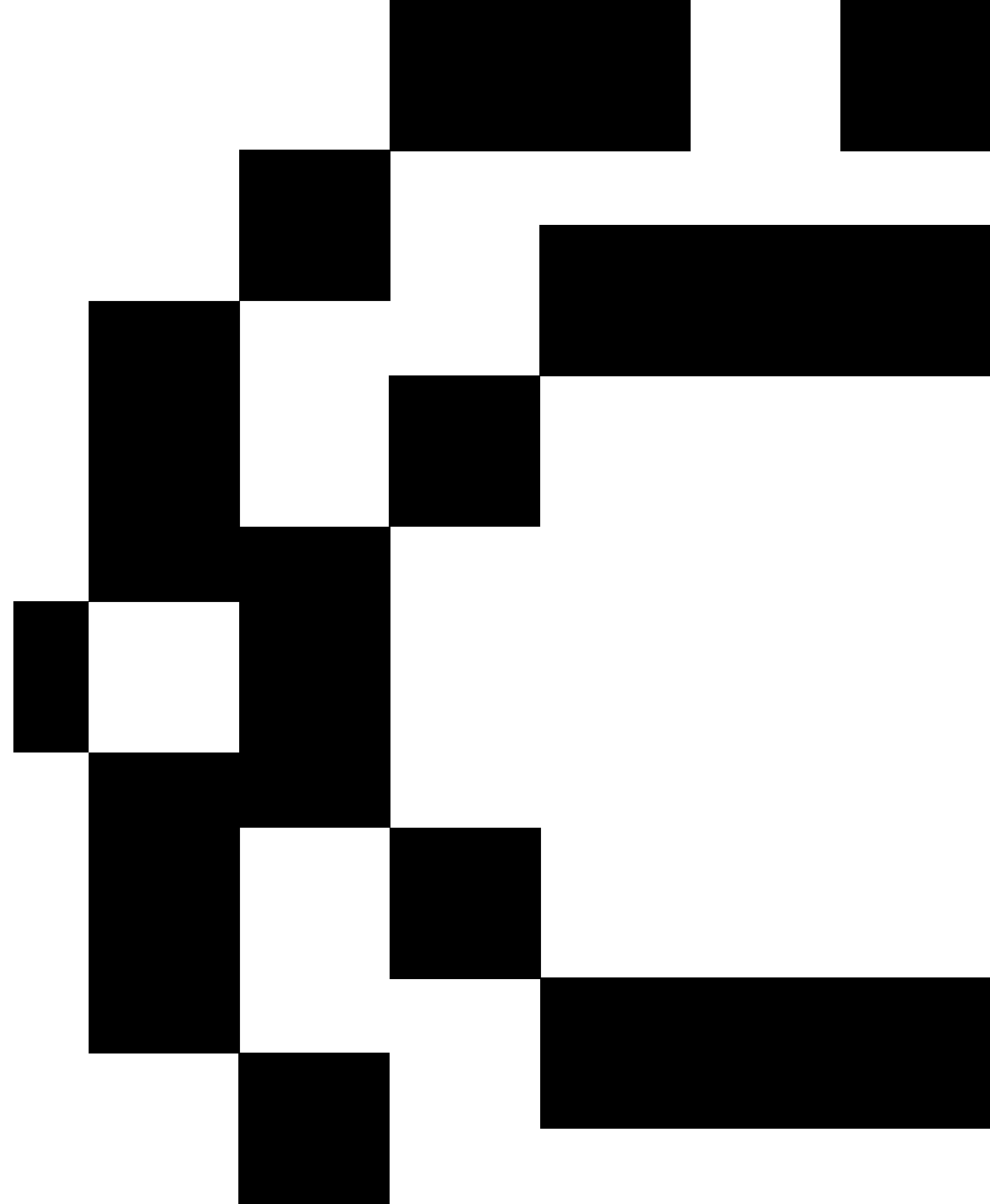


MLOps

Course overview

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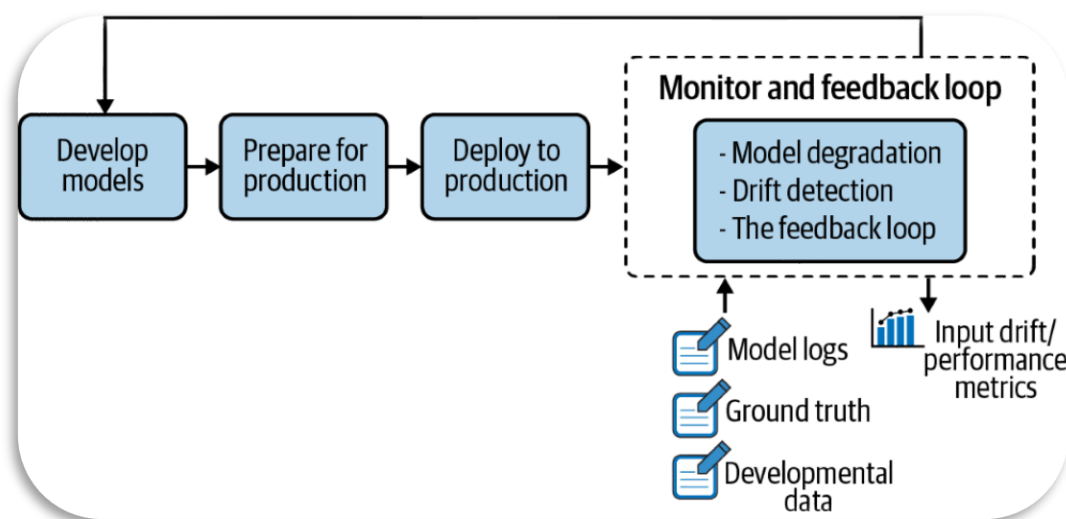




Model Monitoring

Data Drift

04: Monitoring and feedback loop

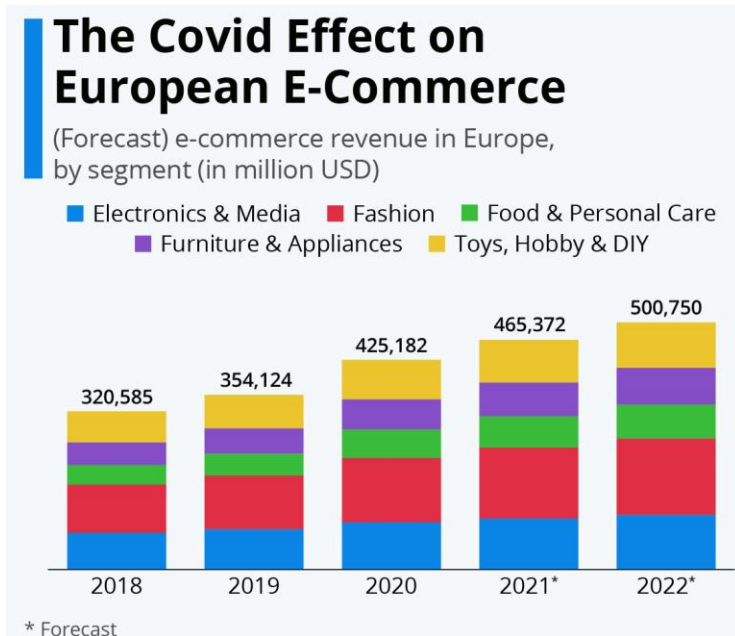


- Machine learning models need to be monitored at three levels:
 1. At the resource level, including ensuring the model is running correctly in the production environment.
 2. At the performance level, meaning monitoring is questioning if the model is still an accurate representation of the pattern of new incoming data or if it is performing as well as it did during the design phase.
 3. Predictions must come with an explanation (i.e., which features have the most influence on the prediction). This kind of information is usually computed with techniques such as Shapley value computation and should be logged to identify potential issues with the model.

Model performance monitoring attempts to track this degradation, and, at an appropriate time, it will also trigger the retraining of the model.

04: Monitoring and feedback loop

- The real world is not perfect!!! Its realism has been challenging Machine Learning perfect assumptions: **data tends to change over the time.**
- Models trained in the past may become less accurate with the time evolution.



- **Since covid, payments changed. Card present transaction are decreasing** (we start to have long "online black fridays").
- Models to predict revenue of merchants had to adapt to this new profiling.
- Recommendation systems and marketing campaigns had to **switch from an on-premise ecosystem to an online prediction and analysis of the customer behaviour.**

04: Monitoring and feedback loop

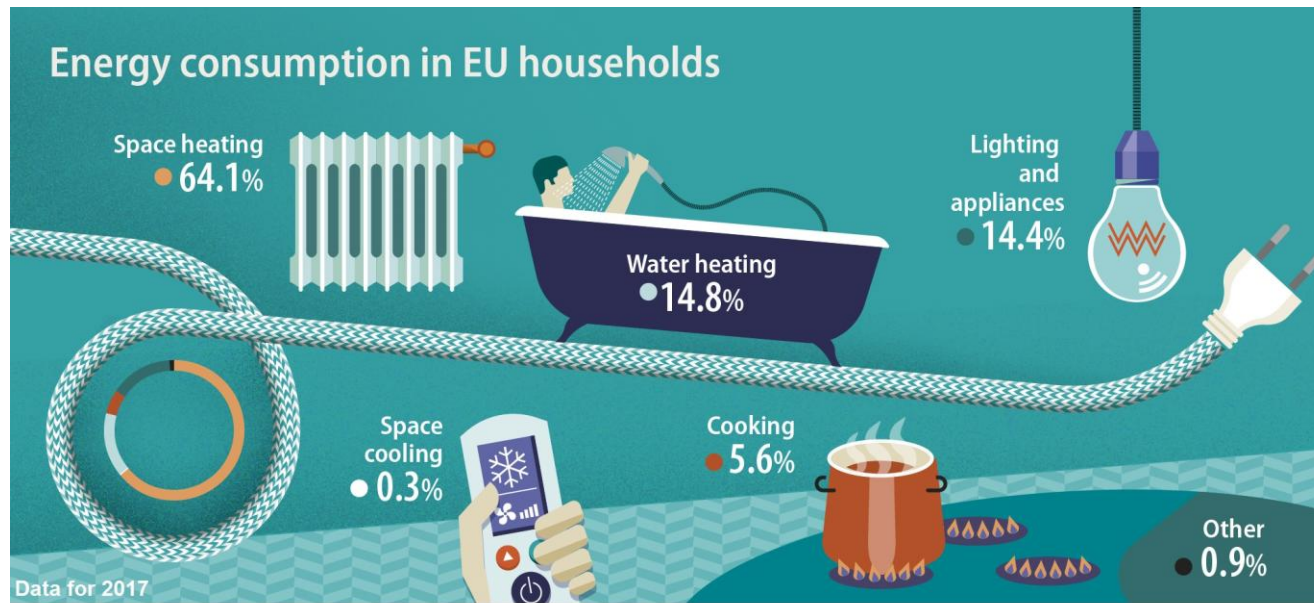
- The real world is not perfect!!! Its realism has been challenging Machine Learning perfect assumptions: **data tends to change over the time.**
- Models trained in the past may become less accurate with the time evolution.



- Scam patterns change with time

04: Monitoring and feedback loop

- The real world is not perfect!!! Its realism has been challenging Machine Learning perfect assumptions: **data tends to change over the time.**
- Models trained in the past may become less accurate with the time evolution.

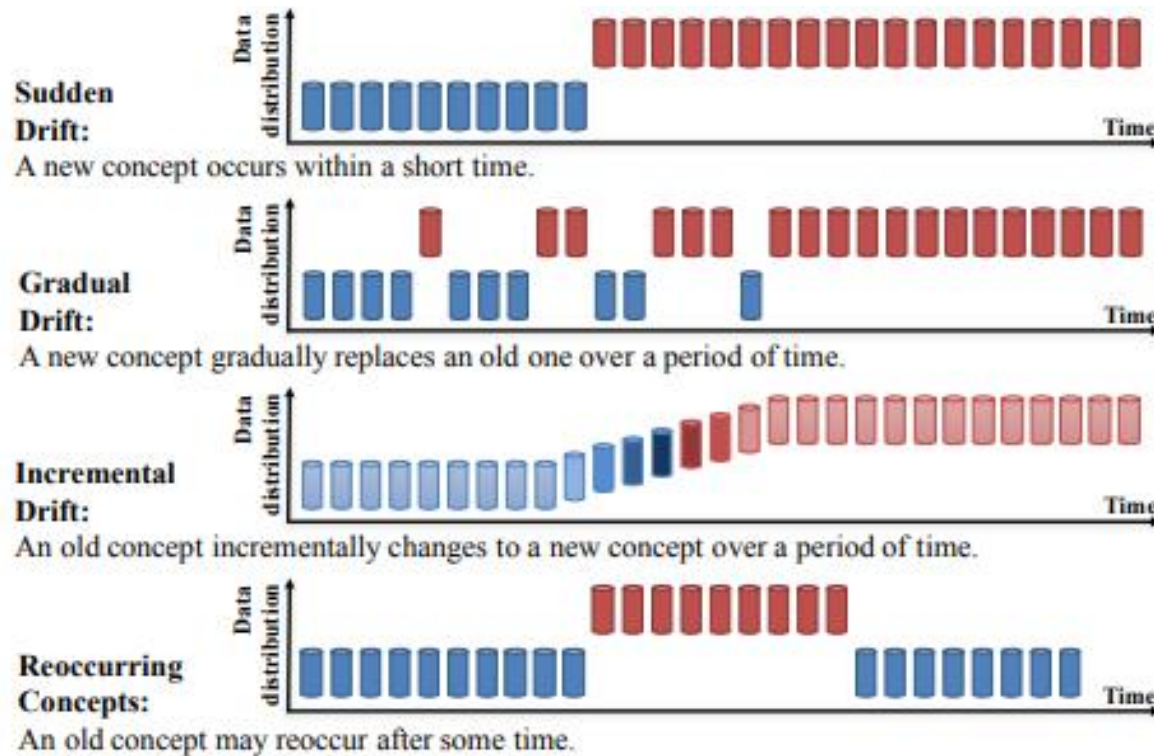


ec.europa.eu/eurostat

- After covid, energy consumption patterns change.
- Remote work challenges seasonality.

04: Monitoring and feedback loop

What is drift in Machine Learning?
Ravages of drift in stream



Sudden drift – A new concept occurs in a short period of time (e.g. at the beginning of COVID-19 in March 2020, stock prices suddenly changed).

Gradual drift – A new concept gradually replaces an old one over an extended period of time (e.g. you witness fewer new oil companies and more new tech companies).

Incremental drift – an old concept incrementally changes to a new concept over some period of time (e.g. a stock price gradually and steadily increases).

Recurring concepts – an old concept may reoccur after some time (e.g. changes in food delivery volumes on the weekends vs weekdays).

Source: <https://arxiv.org/pdf/2004.05785.pdf>

04: Monitoring and feedback loop

What is drift in Machine Learning?
Ravages of drift in stream

Joao Gama



Professor, Faculty of Economics, [University of Porto](#), and INESC TEC
Dirección de correo verificada de [fep.up.pt](#) - [Página principal](#)

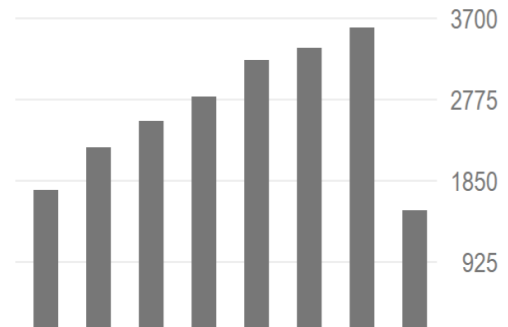
[Data Mining](#) [Machine Learning](#) [Data Stream Mining](#) [Concept Drift](#)

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Total Desde 2019

Citas	29631	17044
Índice h	74	53
Índice i10	275	175



TÍTULO

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AÑO

[A survey on concept drift adaptation](#)

J Gama, I Žliobaitė, A Bifet, M Pechenizkiy, A Bouchachia
ACM computing surveys (CSUR) 46 (4), 1-37

3332

2014

[Learning with drift detection](#)

J Gama, P Medas, G Castillo, P Rodrigues
Advances in Artificial Intelligence–SBIA 2004: 17th Brazilian Symposium on ...

1926

2004

[Learning under concept drift: A review](#)

J Lu, A Liu, F Dong, F Gu, J Gama, G Zhang
IEEE transactions on knowledge and data engineering 31 (12), 2346-2363

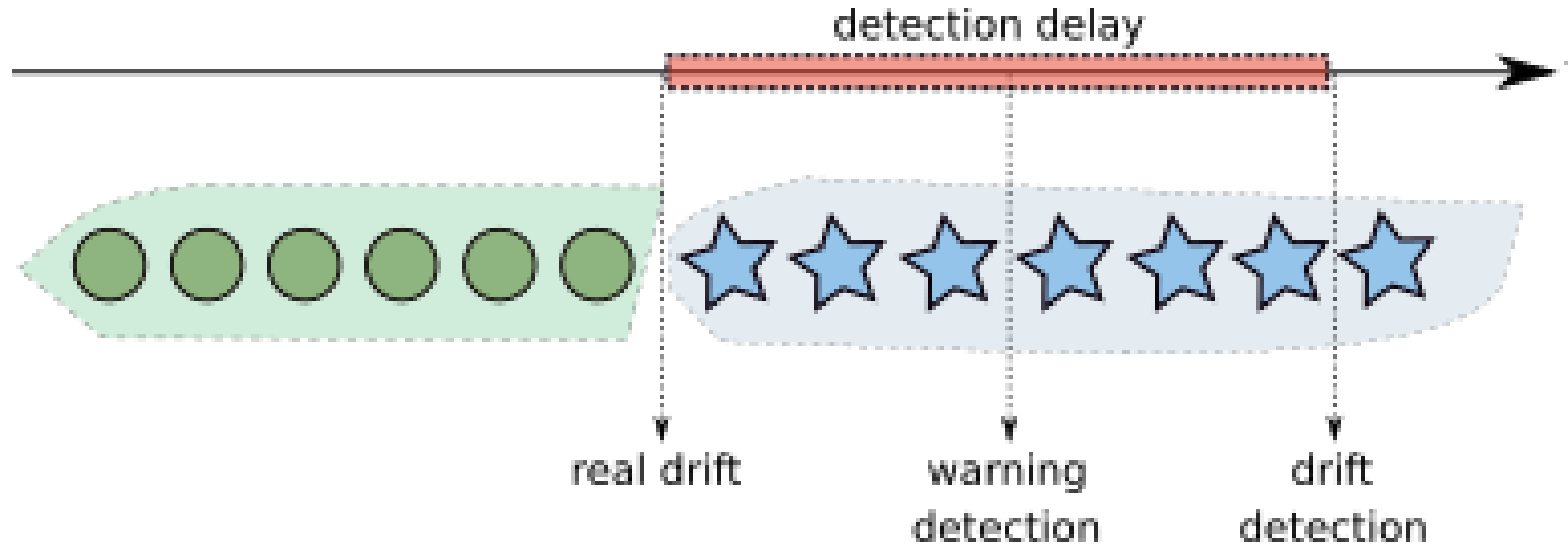
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04: Monitoring and feedback loop

What is drift in Machine Learning?

Drift reaction

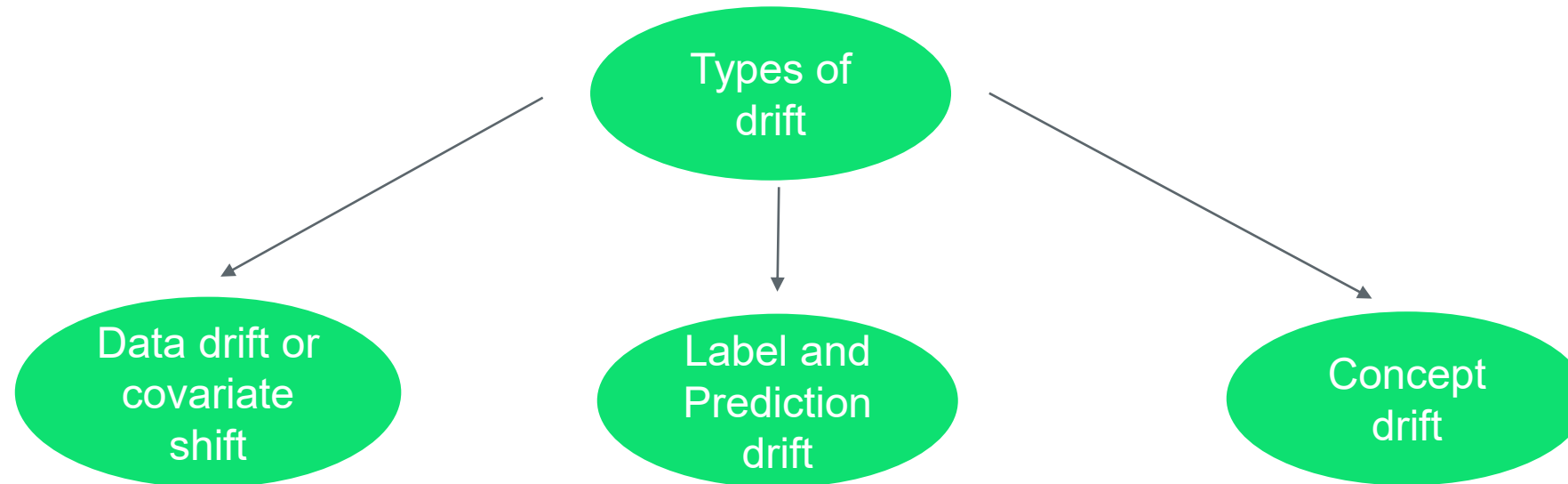


- It is fundamental to have a strong logging system capable of detecting drift.
- The reaction to the drift will depend on the type of drift detected.

04: Monitoring and feedback loop

What is drift in Machine Learning?

- Drift refers to some changes in the data/patterns distribution over a period of time due to:
 1. Changes in the data collection process (**unit data test are a must in MLOps**).
 2. Changes in the data source (**including changing patterns on our target**).
 3. Changes in the business goals.
 4. Sample bias or non stationary environments.

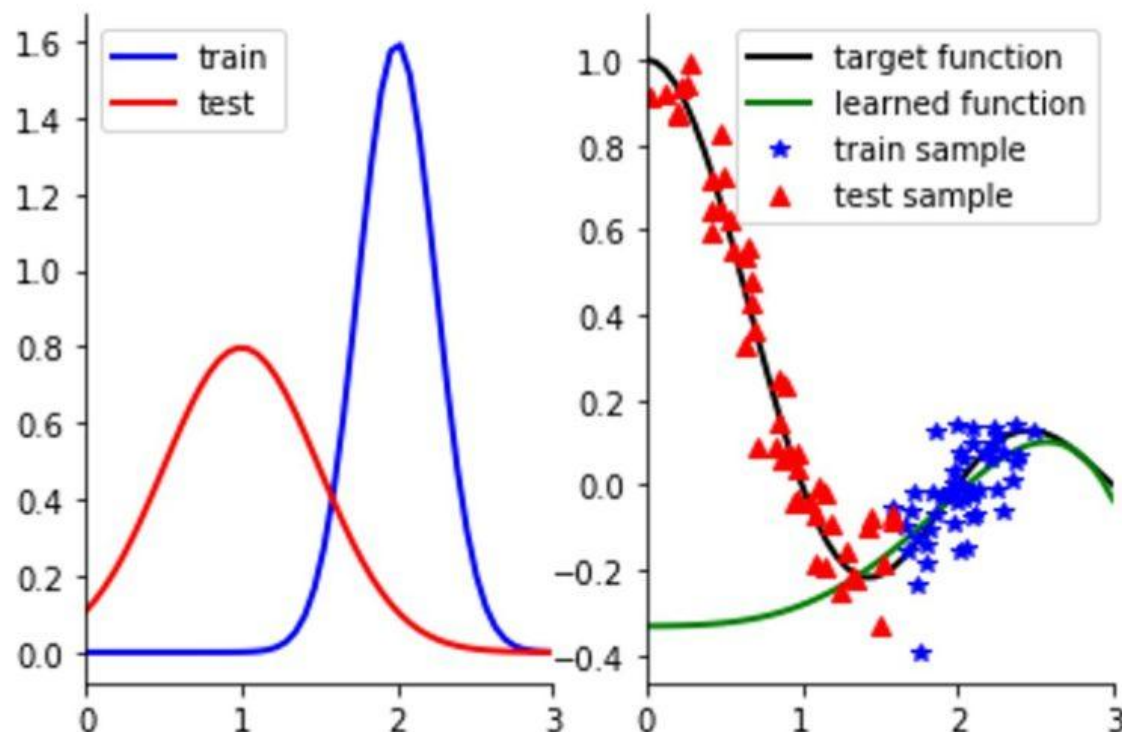


04: Monitoring and feedback loop

Data drift/Covariate shift

- Corresponds to a change on the distribution of the input features used to train the model.
- This could be introducing new values of categorical features, natural and external shifts in data (as the temperature variations across the year), changes on data quality.

Definition: Let be $P(x)$ the marginal probability of the input features x , and $P(y/x)$ the condition probability of the targets y given the input features of x .



$$P_{train}(x) \neq P_{test}(x)$$
$$P_{train}(y/x) = P_{test}(y/x)$$

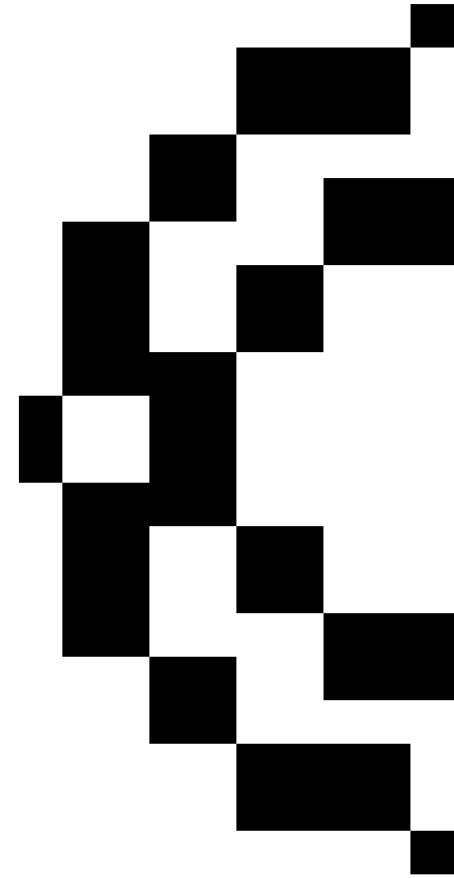
- **Data drift does not mean a change in the relationship between the input variables and the output.**
- **Input distribution shift makes the model less capable of predicting labels.**

04: Monitoring and feedback loop

Data drift/Covariate shift

- Actions that one should take in order to detect data drift:
 1. **Monitor the input data:** If we have a profiling system and unit data test on the input data (Great expectations and Pydeequ) we can easily detect any significant change on the data distribution. Data visualization of the feature distribution can also help.
 2. **Monitor the performance of the model:** any degradation of the model performance can be a consequence of the presence of some data drift.
 3. **Drift detection methods:** statistical tests and specialized algorithms to detect drifts.

The first two steps should be already included in our MLOps framework, from the last classes. **The last one is our novel element in our framework!**



04: Monitoring and feedback loop

Statistical tests

Statistical tests

- **Jensen-Shannon divergence**
- **Two-Sample Kolmogorov-Smirnov (KS)**
- **Kullback–Leibler divergence**
- **Population Stability Index (PSI)**
- One-way Chi-Squared Test
- Two Chi-Squared Contingency Test
- Fisher's Exact Test
- Wasserstein Distance
- Anderson Darling
- Shapiro-Wilk
- Jarque-Bera
- ...

04: Monitoring and feedback loop

Population Stability Index

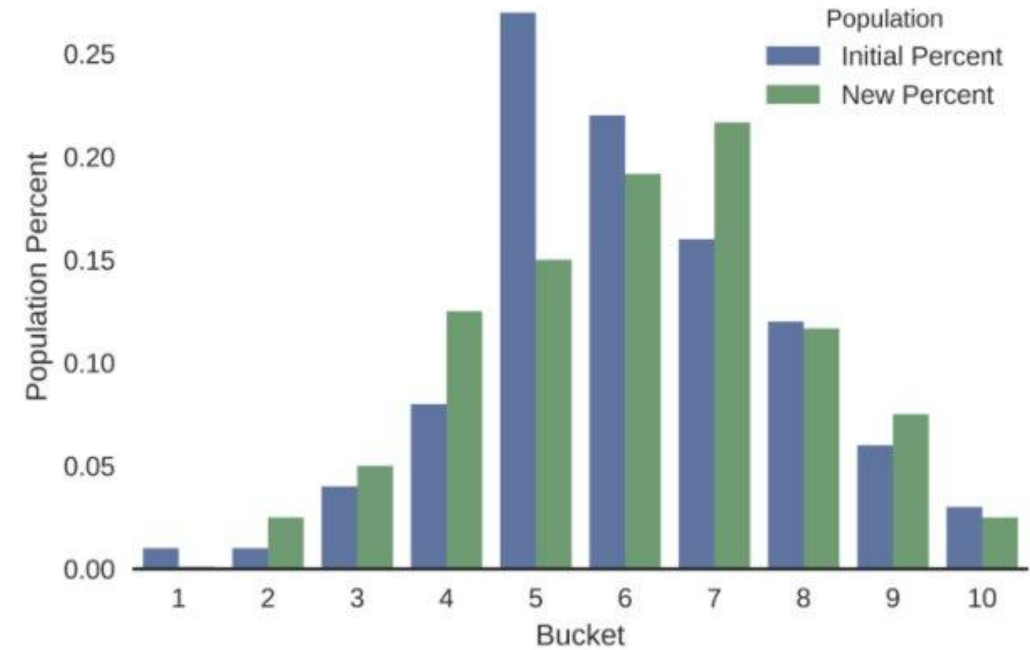
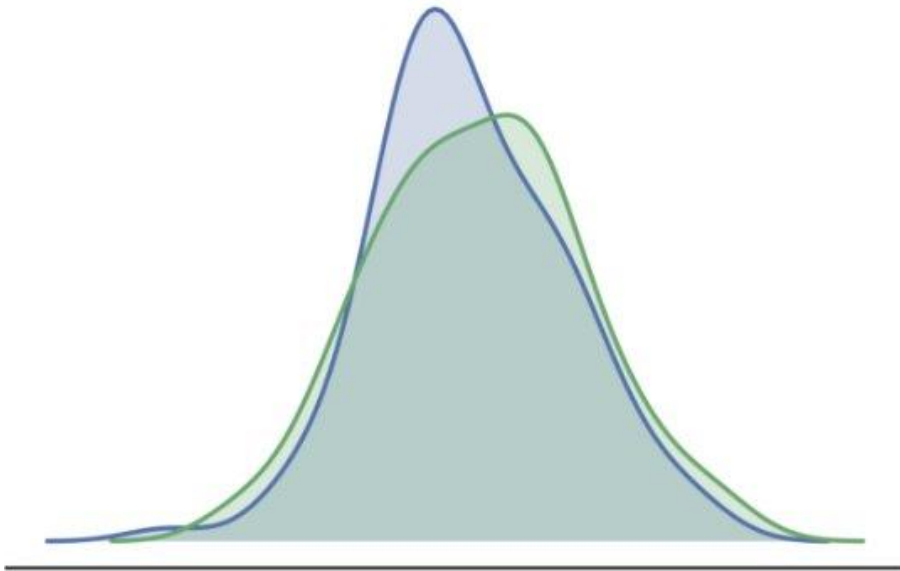
- PSI (Population Stability Index) is a measure of difference between two discrete distributions.
- It is typically used to assess representativity – i.e., assess whether distribution of a binned variable differs in two different data samples which are typically from two different time periods.

$$PSI = \sum_i^n (p_i(x) - q_i(x)) * \log \left(\frac{p_i(x)}{q_i(x)} \right) \quad \text{where } p(x) \text{ is the actual distribution and } q(x) \text{ is the expected distribution.}$$

- It is widely used in credit risk score cards to monitor changes in the variable distributions, banks rating.
- We need always a reference data set that should be used to compare with the production data.
- PSI can be used for finding unstapled features, for model validation and monitoring if we choose the score as the variable to analyze.

04: Monitoring and feedback loop

Population Stability Index



04: Monitoring and feedback loop

Population Stability Index

- We can create a generic receipt to build our PSI indicators:
 1. Divide the numeric attributes into the specified number of intervals or bins.
 2. Calculate % of records in each interval based on modeling sample.
 3. Obtain the cutoff points for all intervals.
 4. Apply the cutoff points to a different sample, usually a validation sample.
 5. Generating the distributions for the validation sample.
 6. Sum up the statistics across all intervals for both modeling and validation samples to calculate PSI.

PSI value	Conclusion
< 0.1	No significant population change
<0.2	Moderate population change
>= 0.2	Significant population change

04: Monitoring and feedback loop

Population Stability Index

- PSI is not symmetric, do not switch the order of the reference and production data.
- Binning is important: when considering continuous variables, the number of bins affect PSI values. Some exploration is required, just to find the optimal number of bins.
- Don't be afraid of double-checking PSI values with a visual component of a plot with the respective distributions.
- PSI has a problem with 0 probability events (remember the formula $\log(\text{actual/expected})$).

Final challenge: Can I use PSI for feature selection ?
What would be the implications?

04: Monitoring and feedback loop

KL(Kulback-Leibler) divergence

- To detect data drift, we can be inspired by information theory and pick the so called entropy: **which is a reflection of the amount of information need for removing the uncertainty.**
- Let q be the estimated probability distribution and p the true probability distribution.

$$D_{KL}(p||q) = - \sum_i^n (p_i(x)) * \log \left(\frac{q_i(x)}{p_i(x)} \right)$$

Case of discrete variable x

$$D_{KL}(p||q) = - \int (p(x)) * \log \left(\frac{q(x)}{p(x)} \right) dx$$

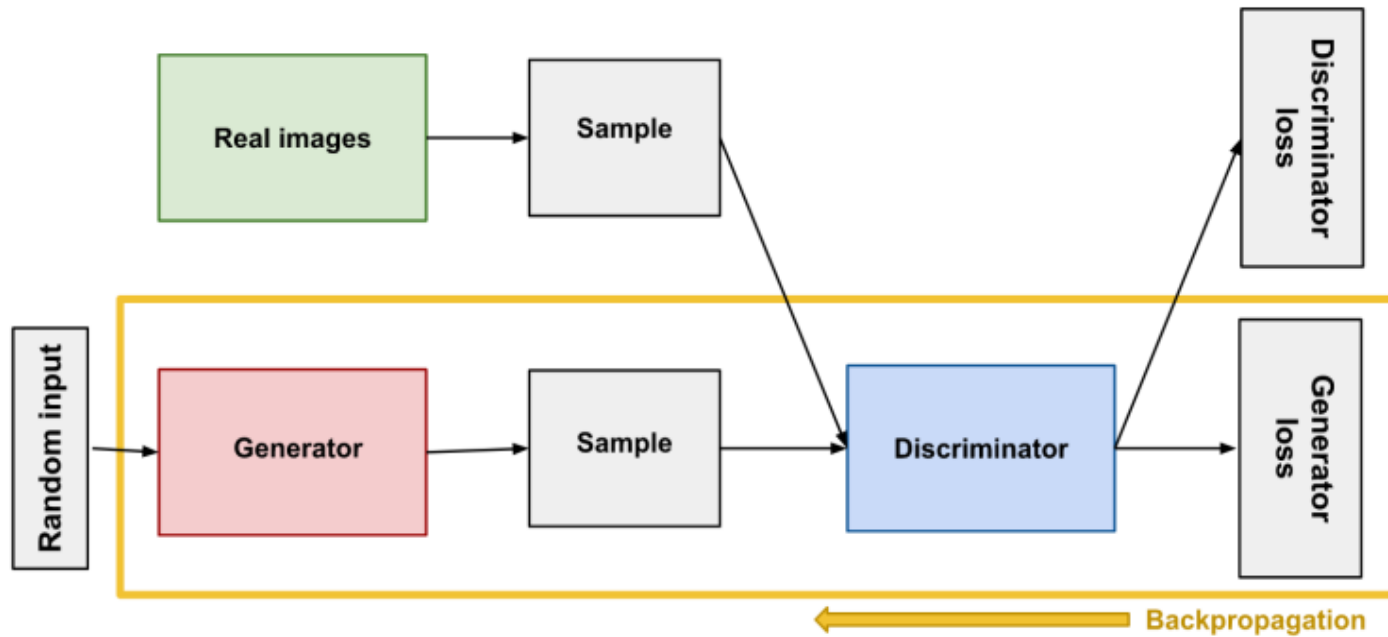
Case of continuous variable x

- The Kullback–Leibler can be thought as **a measure of entropy increase due to the use of an approximation to the true distribution rather than the true distribution itself.**
- It is not symmetric. The divergence between q and p is different from p and q , so this metric is not a true distance metric.
- Issues dealing with null probability events.

04: Monitoring and feedback loop

KL(Kulback-Leibler) divergence

If ... KL divergence measures the inefficiency caused by the approximation...



Useful for GANs
(Generative Adversarial
Networks)



Synthetic data!!!!

04: Monitoring and feedback loop

JS(Jensen-Shannon) divergence

- We can call it as information radius or total divergence of the average. **It is a statistical metric based in information theory and measures the relative entropy between two distributions.** Let q and p be two different probability distributions:

$$D_{JS}(p||q) = \frac{1}{2} D\left(p||\frac{p+q}{2}\right) + \frac{1}{2} D\left(q||\frac{p+q}{2}\right) \text{ where } D(p||q) = \int p(x) \log\left(\frac{p(x)}{q(x)}\right) dx$$

In model monitoring, the discrete form of JS divergence is typically used to obtain the discrete distributions by binning data

Case of continuous variable x

$$\text{or } D(p||q) = \sum p_i(x) \log\left(\frac{p_i(x)}{q_i(x)}\right)$$

Case of discrete variable x

- The advantage of JS divergence over other metrics **like is mostly related to issues with empty probabilities for certain events or bins.**
- The Jensen-Shannon divergence is a symmetrized version of the Kullback-Leibler divergence**, which allows it to be used to compare any two probability distributions, regardless of which distribution is considered the “reference” and which is considered the “comparison.”

04: Monitoring and feedback loop

JS(Jensen-Shannon) divergence

- JS divergence is typically applied to each feature independently.
- It is not designed as a covariant feature measurement but rather a metric that shows **how each feature has diverged independently from the baseline values**.
- It's not a multi-variate drift measurement $\longrightarrow D_{JS}(p||q) = \frac{1}{2} D\left(p||\frac{p+q}{2}\right) + \frac{1}{2} D\left(q||\frac{p+q}{2}\right)$
- The traditional receipt for this kind of tests **is start by binning data**:
 1. Find a robust method (taking into consideration a certain scale of features and data).
 2. Create mixture distribution for comparison using the production and baseline distributions. This makes JS divergence a symmetric metric (such as PSI). 0 bins are handled naturally (mixture terms of distributions in log).

$$\sum p(x) \log \left(\frac{2 p(x)}{q(x) + p(x)} \right) + \sum q(x) \log \left(\frac{2 q(x)}{q(x) + p(x)} \right) \rightarrow \sum q(x) \log (2)$$

(0*ln(0) = 0) as the function is smooth and has a limit as it approaches 0

3. Compare production and baseline to mixture.

04: Monitoring and feedback loop

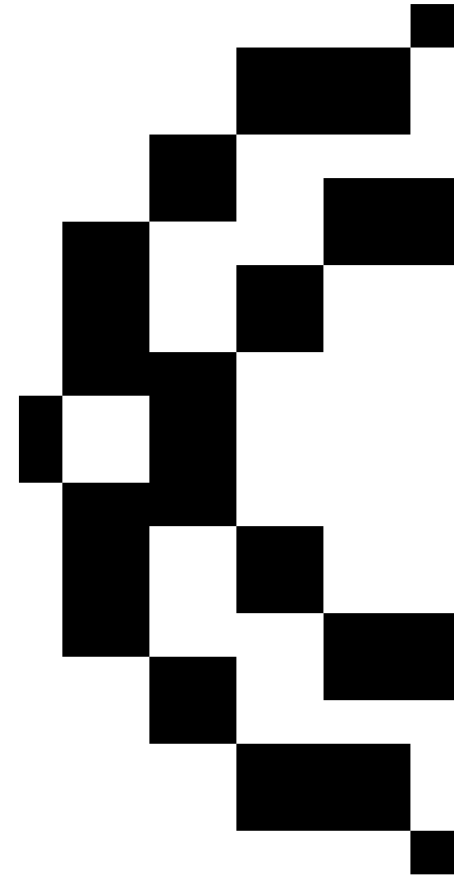
KS(Kolmogorov-Smirnov) test

- We can use KS test **as a nonparametric statistical test**, therefore does not make any assumptions about the underlying distribution.
- **The null hypothesis is that the two samples come from the same distribution.** The KS test is applied to reject or accept it. If the null is rejected then we can conclude that there is a drift in the model.
- We can use it to test whether two underlying one-dimensional probability distributions differ.

$$D = \sup_x |F_1(x) - F_2(x)|$$

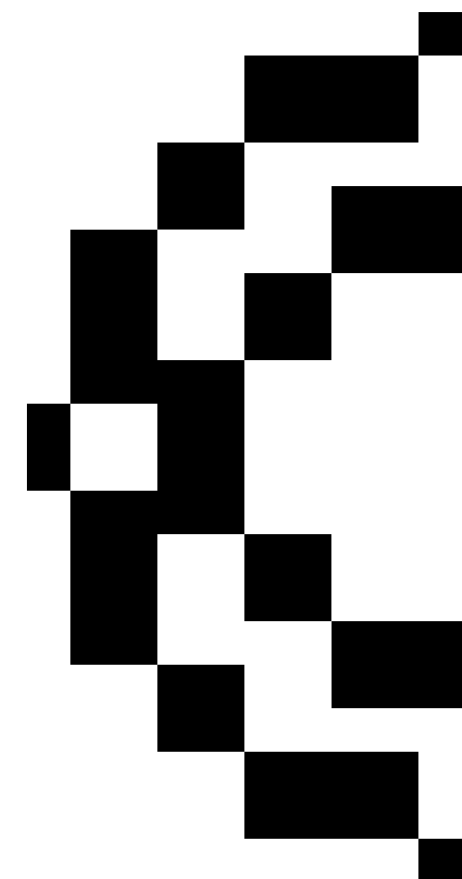
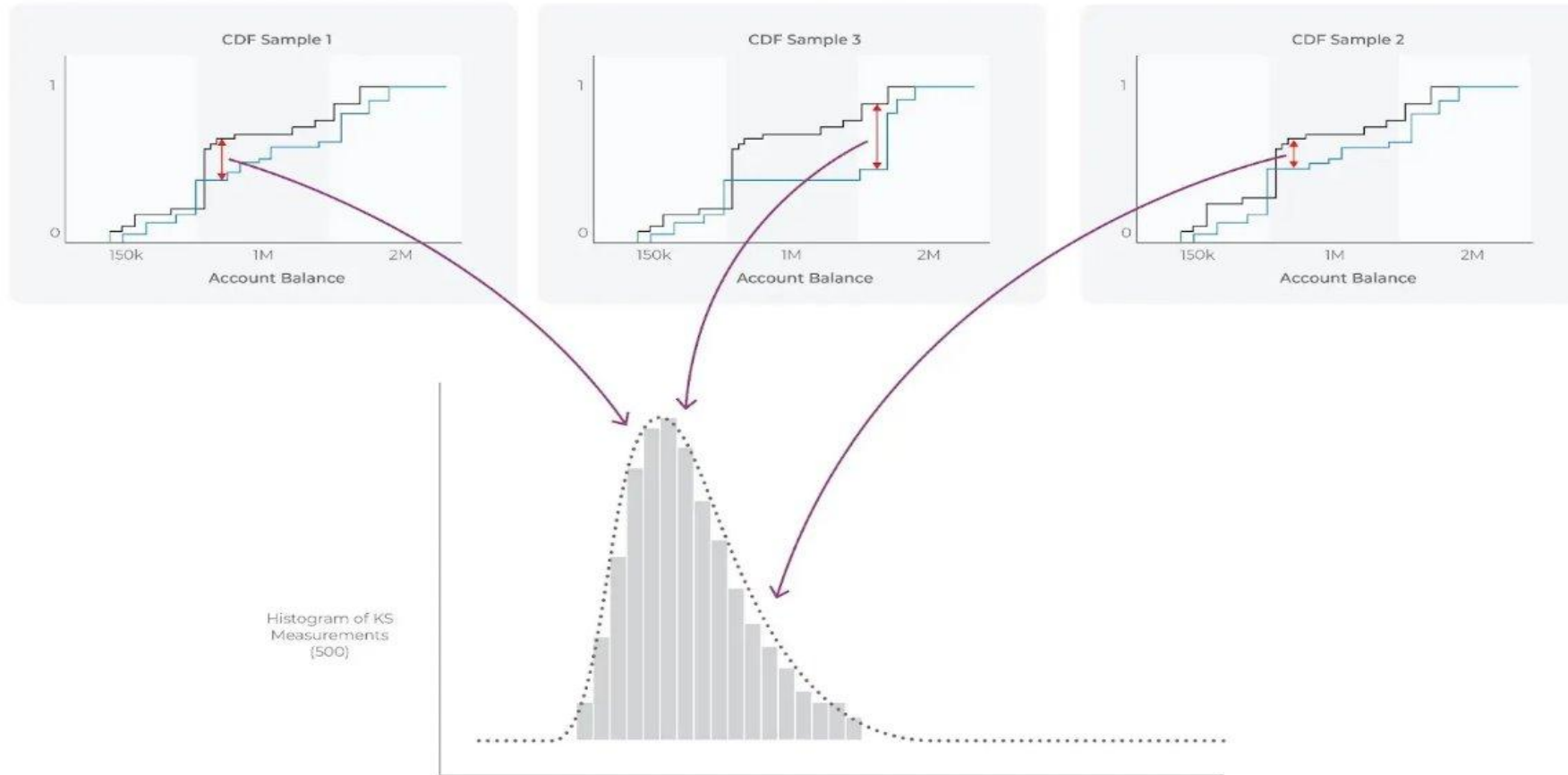
F's are cumulative distribution functions of the samples. We compare the result with the critical values from the table with the Kolmogorov distribution.

- It is like a distance between the empirical distribution function of the sample and the cumulative distribution function of the reference distribution.
- At production level we will take several samples to compare, so we will have several values for the KS test. This means that we can generate a distribution with these values - Kolmogorov distribution. **Here we will apply the p value concept.**



04: Monitoring and feedback loop

KS(Kolmogorov-Smirnov) test



04: Monitoring and feedback loop

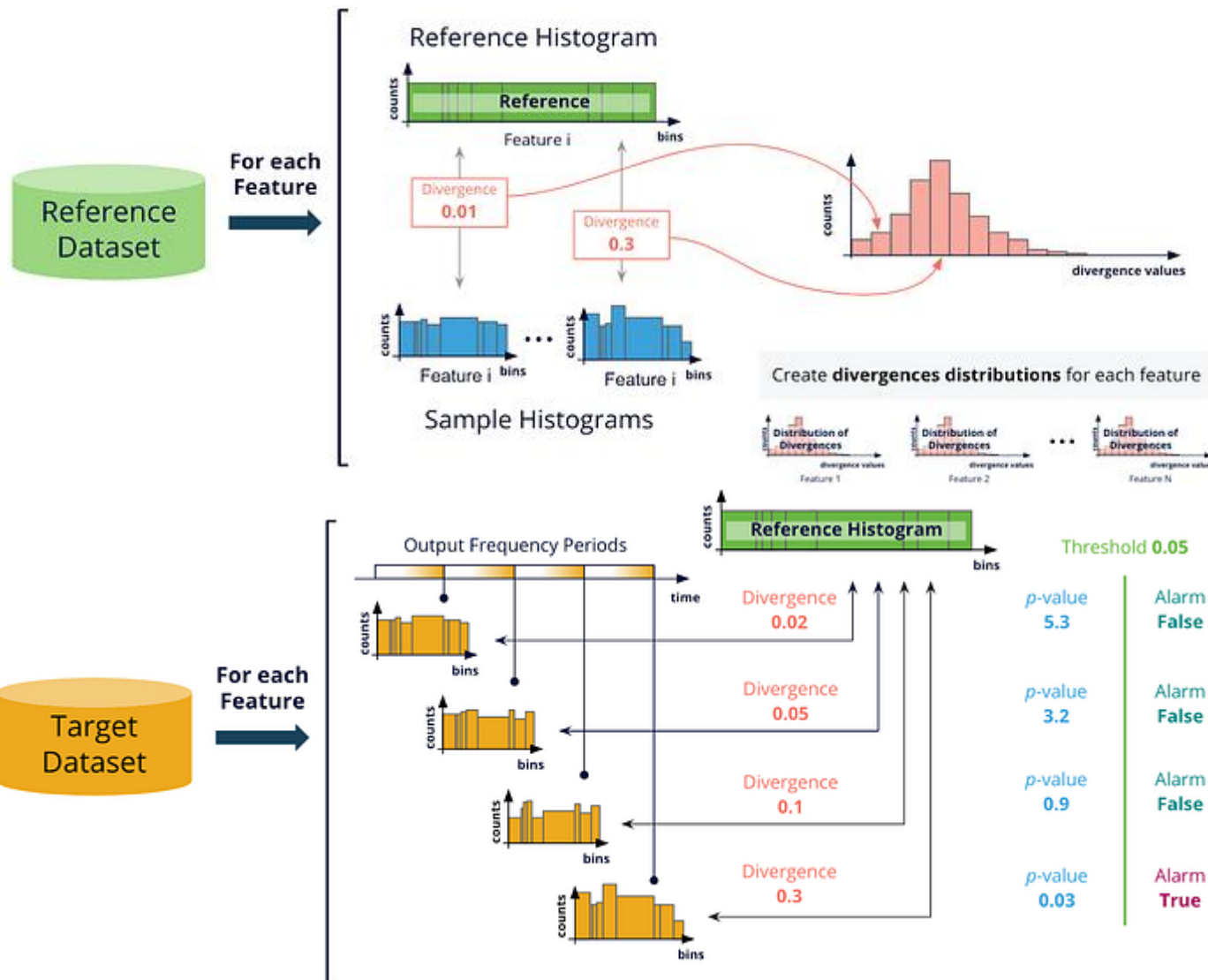
KS(Kolmogorov-Smirnov) test

- Limitations:
 1. It only applies to continuous distributions.
 2. It tends to be more sensitive near the center of the distribution than at the tails: it is much harder for the data to deviate that much in the tail, and KS evaluates the differences in the tail and in the center with "the same weights". **It is insensitive to how much of a difference there is.**
 3. The distribution must be fully specified. That is, if location, scale, and shape parameters are estimated from the data, the critical region of the KS test is no longer valid (**one can take wrong conclusions**). It typically must be determined by simulation.

Information content can change between two distributions, but a KS test statistic may not change. On the other hand the information theory-based metrics, any movement in data between bins is captured in a metric change.

04: Monitoring and feedback loop

Resume

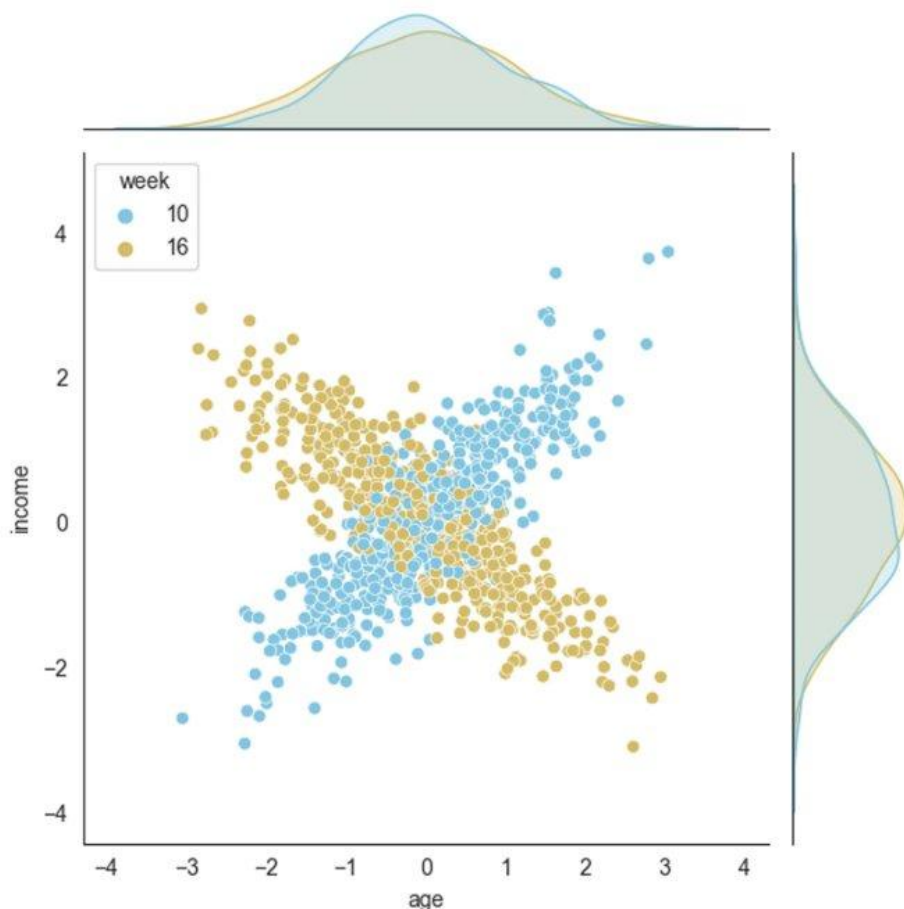


- Update the feature histogram in a new time step and **compare with the Reference histogram to compute divergence value.**
- **Compare the p-value** to be within the expected distribution of divergences is computed.
- After all p -values have been calculated, a multivariate hypothesis test is applied under **the null hypothesis that the divergence values observed for all the features in the Target data follow the same distributions as in the Reference data.**

04: Monitoring and feedback loop

When does univariate drift detection fail?

- The applied statistical tests **are univariate!** Each feature is treated in isolation from the rest.



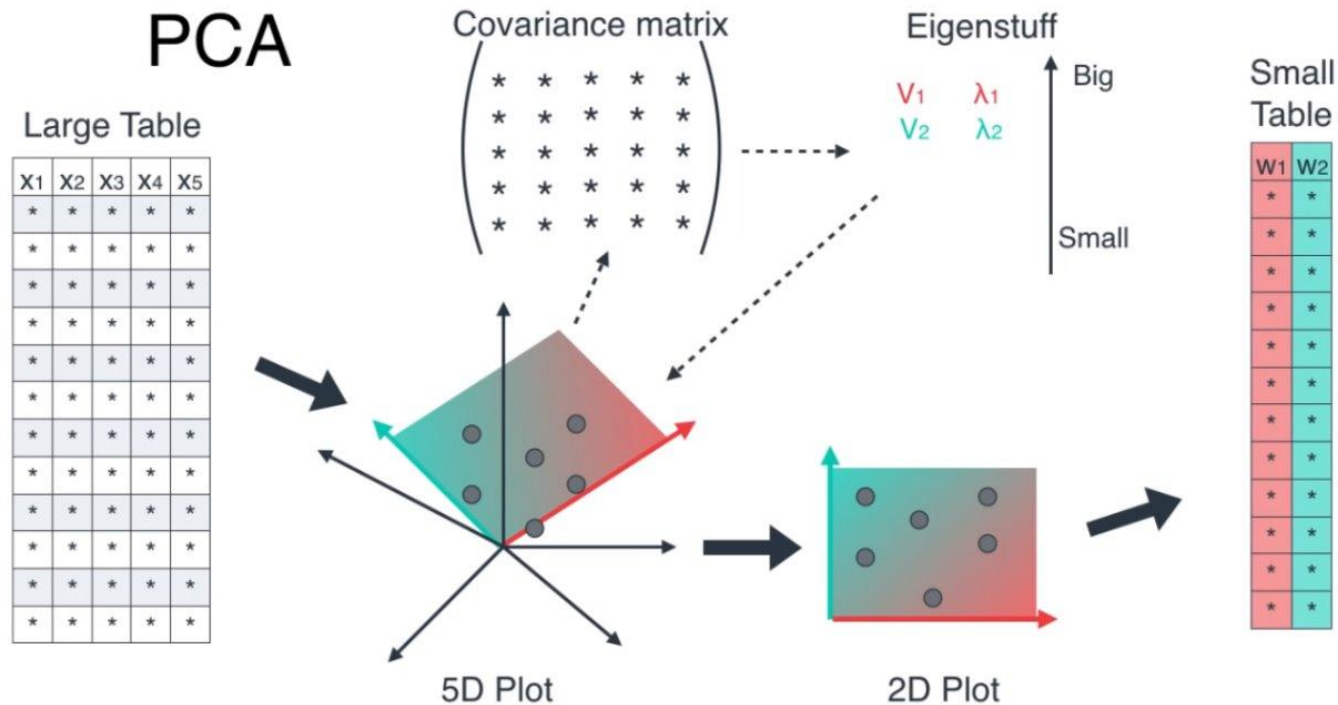
- In the 10th week (blue), there is a strong positive correlation between the two features.
- In the 16th week (yellow), the marginal univariate distribution of each feature is still standard normal as before but are negatively correlated.



Strong covariate shift
No change on the distribution of the features
Need of a multivariate approach that can capture the changes in the joint features.

04: Monitoring and feedback loop

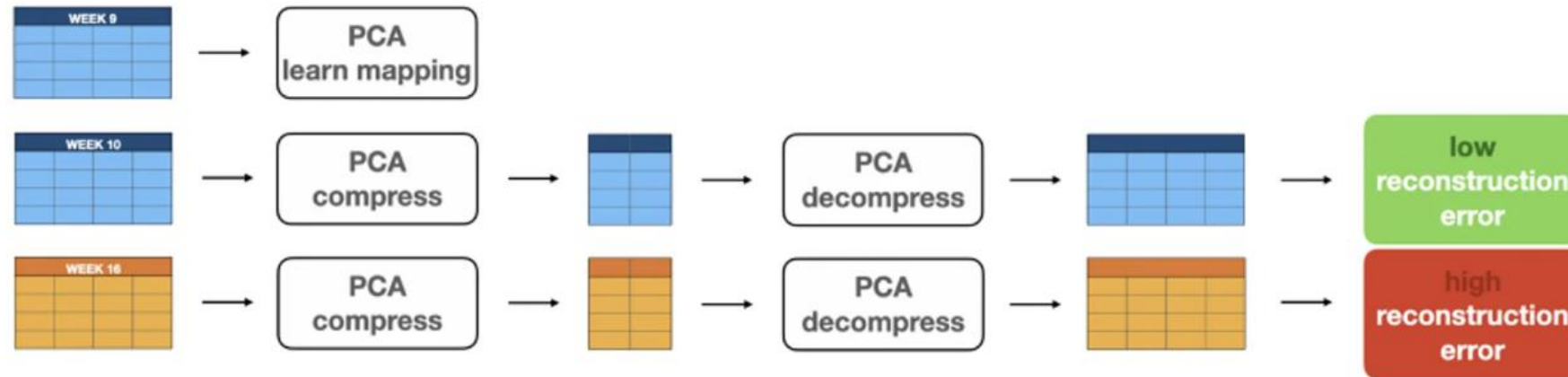
Multivariate approach



1. Dimensionality reduction by identifying the directions of our data that contain most of the variance.
2. Data reconstruction using the eigenvectors and the transformed data.
3. Estimate the expected reconstruction error: **the result will not be the same as the original data due to the lossy compression process**

04: Monitoring and feedback loop

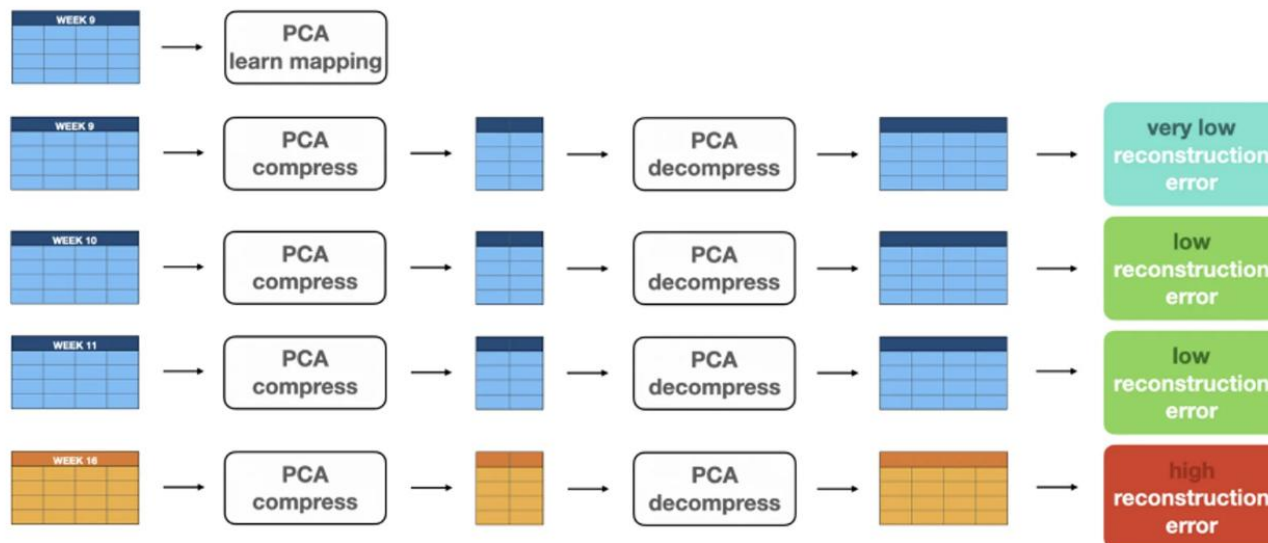
Multivariate approach



- Choose a reference period to learn the PCA mapping - week 9.
- Use the learned mapping to compress and decompress week 10 and week 16.
- Week 16 presents a high reconstruction error - covariate shift detection.

04: Monitoring and feedback loop

Multivariate approach issues



Important:

1. Presence of non linear covariate shift, when drift left features' means and standard deviations, as well as the correlation between the features, intact.
2. Stability of the reconstruction error, means no covariate shift

Expected reconstruction error:

- Take a **sample of the training data that we know not to exhibit covariate shift**, our reference dataset.
- **Split it into chunks** and compute the reconstruction error on each chunk.
- **Compute the mean error and its variance.** Check whether the reconstruction error from the data in production is outside of the ranges obtained from the reference set (e.g., use a threshold for standard deviations) which signs for covariate shift.



Model Monitoring

Label Drift

04: Monitoring and feedback loop

Label and Prediction Drift

- Corresponds to **a change on the labels, categorical feature of output data or distribution of the model predictions.**
- It is quite similar to data drift, which means that it is mostly influenced by external factors. Therefore we can use similar strategies for detection.
- The main factors are changes in human judgment, introduction of new categories,...



$$P_{train}(y) \neq P_{test}(y)$$
$$P_{train}(y/x) = P_{test}(y/x)$$

1. Let's consider the plot as a house price prediction, dependent on the house size.
2. The price prediction increased with less larger size houses available in the market. But the relationship between size and price still holds.

- An other example for label drift is customer sentiment analysis based on customer reviews. If the criteria to determine whether a review is positive or negative changes, then it causes label drift.

04: Monitoring and feedback loop

Label and Prediction Drift

- For comparing the output variable, we can always use the previous tests.
- Nevertheless there exists others more focused on the ML target output, as they are "algorithm" driven:

Extra statistical testes

The Page-Hinkley test

ADWIN (Adaptative Windowing)

DDM(Drift Detection Method)

More details:

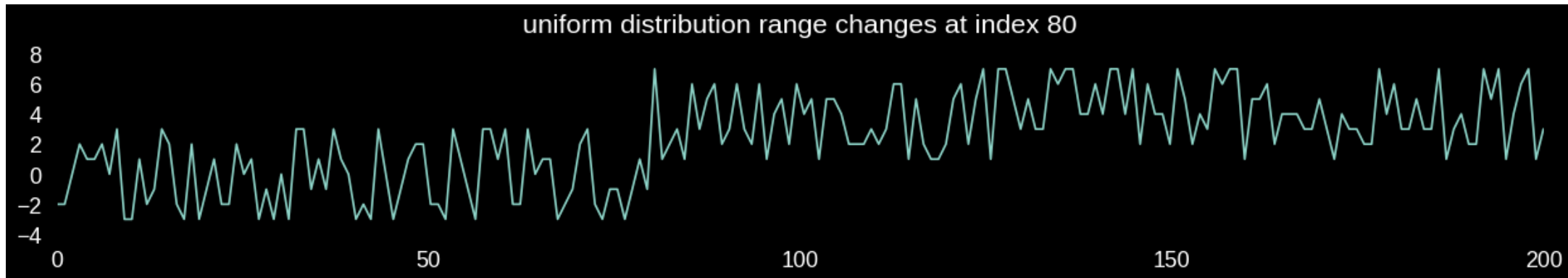
https://link.springer.com/chapter/10.1007/978-3-540-28645-5_29

Challenge for home: explore Hellinger distance as a test for drift. Check for which kind of drift it is useful.

04: Monitoring and feedback loop

Label and Prediction Drift

- The Adaptive Windowing (ADWIN) algorithm **uses a sliding window approach to detect drift**. Window size is fixed and ADWIN slides the fixed window for detecting any change on the newly arriving data.
- **When two sub-windows show distinct means in the new observations the older sub-window is dropped.**
- A user defined threshold is set to trigger a warning that drift is detected. **If the absolute difference between the two means derived from two sub-windows exceeds the pre-defined threshold, an alarm is generated.**



Useful for signal analysis, time series...



Model Monitoring

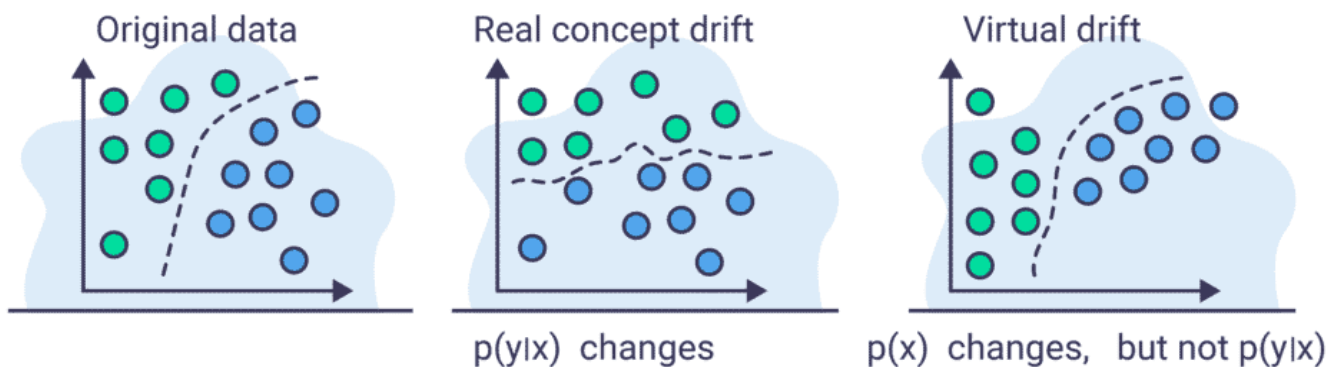
Concept Drift

04: Monitoring and feedback loop

Concept Drift

- **Concept drift is different from covariate shift and label drift:** it is not related to the data distribution or the class distribution but instead is related to the **relationship between both**.
- It is mainly a result of changes in the real-world phenomena being modeled, nevertheless also can be explained by temporary changes that occur in the data that is being collected, or a bias sample in the training process.

Definition: Let be $P(y/x)$ the condition probability of the target label y given the input features of x .

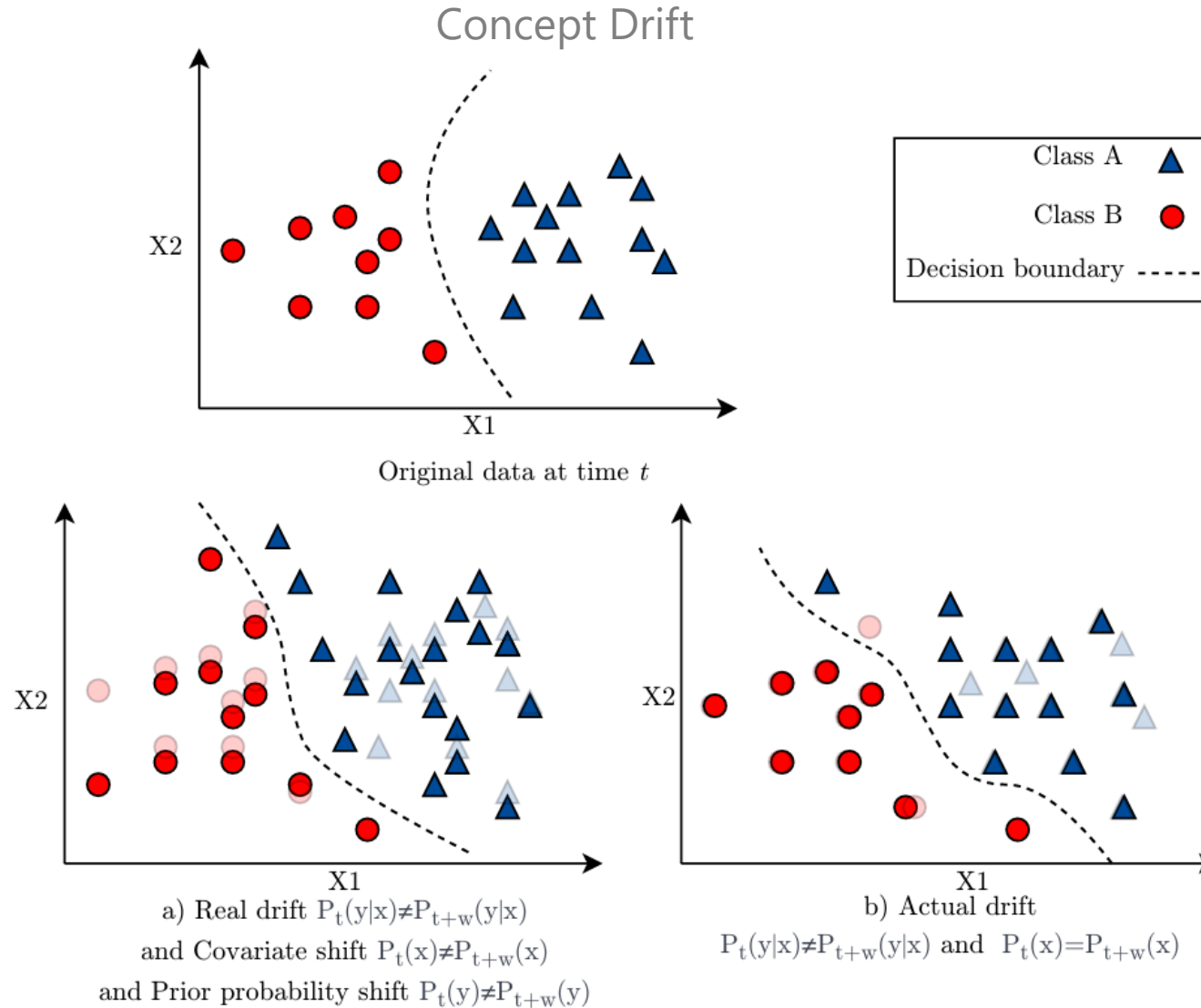


$$P_{train}(y/x) \neq P_{test}(y/x)$$

- The distribution of the input features or the output can change as well, but our focus is on the relationship between them.

Nice paper about concept drift
<https://arxiv.org/pdf/1511.04816.pdf>

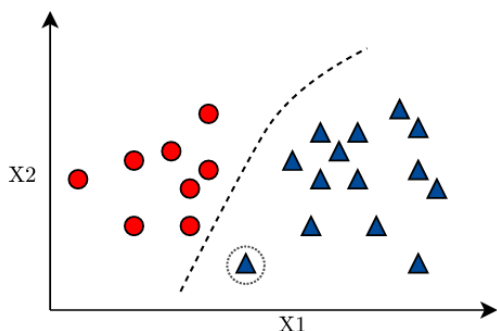
04: Monitoring and feedback loop



<https://arxiv.org/pdf/2203.11070.pdf>

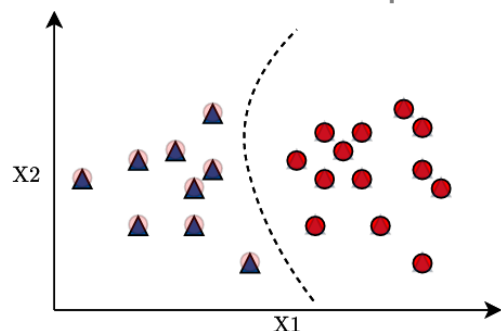
04: Monitoring and feedback loop

Concept Drift



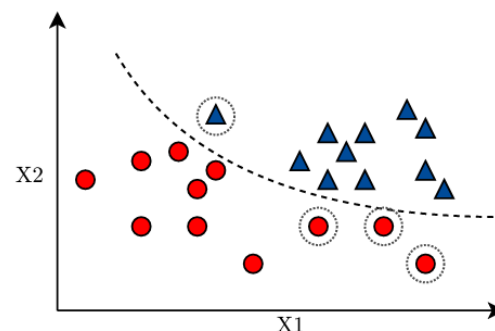
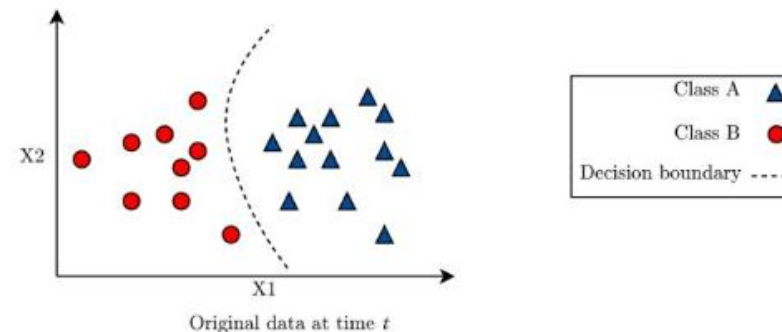
c) Fickle concept drift

$$\exists x(\text{argmax } P_t(y|x)=c_1 \text{ and } \text{argmax } P_{t+w}(y|x)=c_2)$$



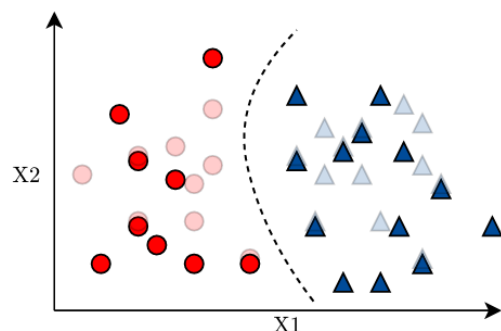
d) Severe concept drift

$$\forall x(\text{argmax } P_t(y|x)=c_1 \text{ and } \text{argmax } P_{t+w}(y|x)=c_2)$$



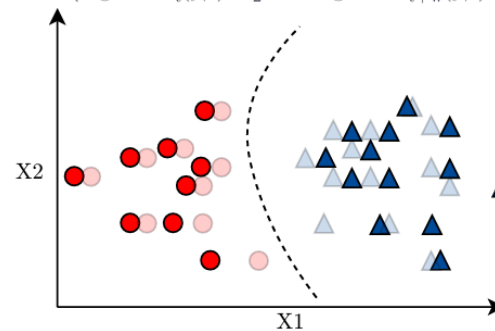
e) Intersected concept drift

$$\exists x(\text{argmax } P_t(y|x)=c_1 \text{ and } \text{argmax } P_{t+w}(y|x)=c_2) \text{ and } \exists z(\text{argmax } P_t(y|z)=c_2 \text{ and } \text{argmax } P_{t+w}(y|z)=c_1)$$



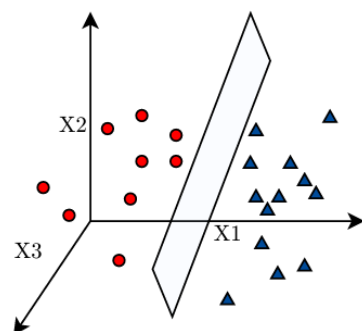
f) Virtual drift

$$P_t(y|x)=P_{t+w}(y|x) \text{ and } P_t(x) \neq P_{t+w}(x)$$



g) Local Concept Drift

$$P_t(X_1) \neq P_{t+w}(X_1) \text{ and } P_t(X_2)=P_{t+w}(X_2)$$



h) Feature evolution

$$X_t=(X_1,X_2) \text{ and } X_{t+w}=(X_1,X_2,X_3)$$

- **Fickle Concept Drift:** When data samples can belong to different classes at different times.
- **Severe Concept Drift:** Involves a complete change in the target classes of all data samples after the drift.
- **Intersected Concept Drift:** Occurs when only a subset of data samples changes their target classes after the drift.
- **Local Concept Drift:** Targets only a sub-region of the feature space, affecting a specific subset of the data.
- **Feature Evolution:** Arises when new attributes dynamically emerge in the input space, altering the feature set.

04: Monitoring and feedback loop

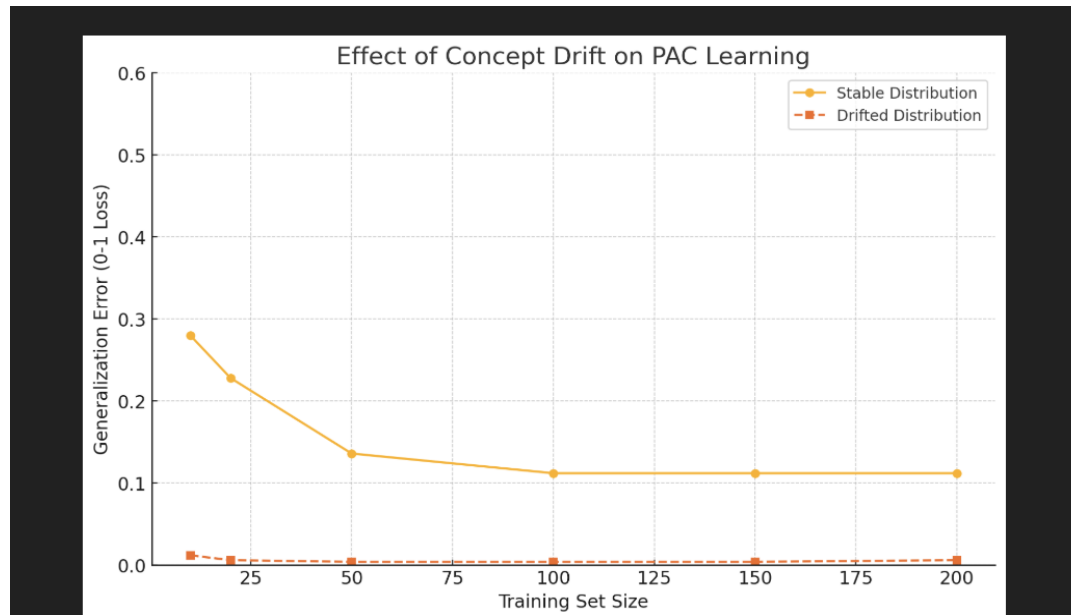
Detecting Concept Drift

Data distribution-based detection

- Data distribution-based detectors use distance measures to estimate the similarity between the data distributions in two different time-windows. Concept drift is detected if the two distributions are significantly distant.

Performance-based detection

- Inspired by PAC (Probably Approximately Correct) framework. A PAC learner tries to learn a concept (by selecting a hypothesis from a set of hypotheses that has a low generalization error). The learner's error rate reduces as the number of examined samples increases as long as the data distribution remains constant.



04: Monitoring and feedback loop

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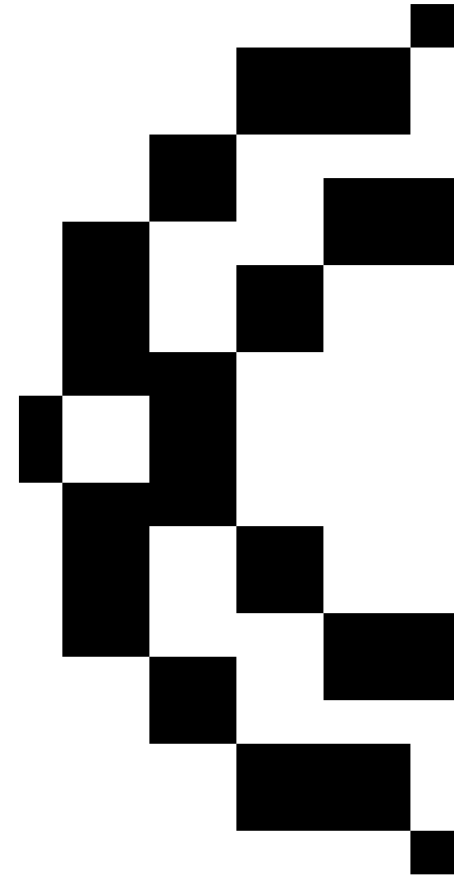
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Multiple hypothesis based detection

- Are hybrid approaches that apply several detection methods and aggregate their results in parallel or hierarchically.

Contextual-based detection

- Contextual-based detectors use context information available from the system and data to detect the drift. We can use explainability to detect concept drift.



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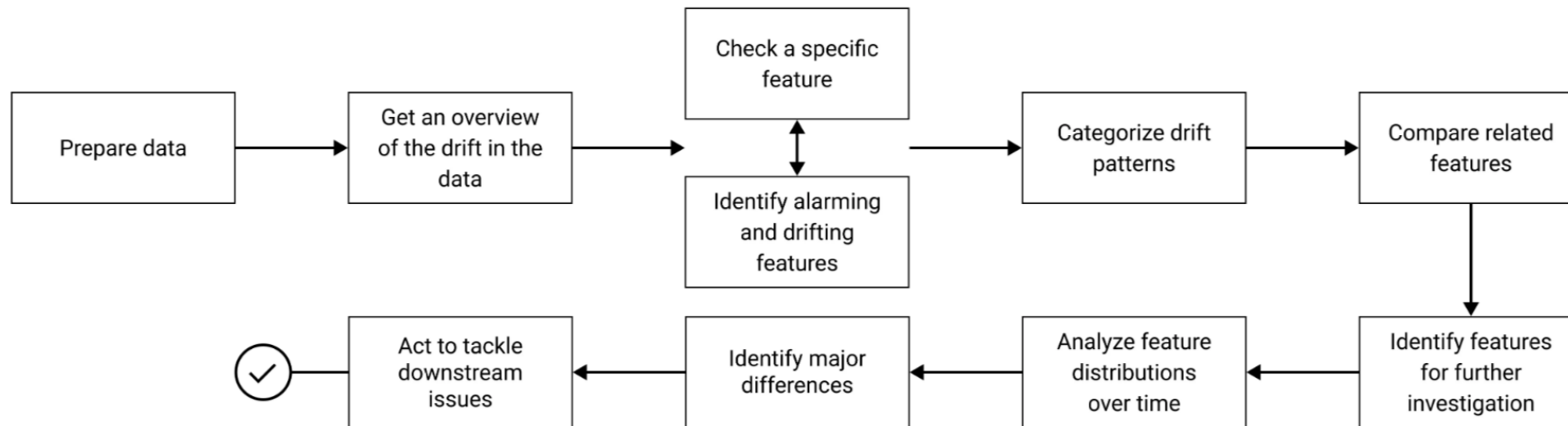
Handling Concept Drift

- We have some strategies to deal with concept drift:
 1. A passive one; the model is updated continuously every time new data instances are received. This one is **quite effective with gradual drifts and recurring concepts**. We can include here online learning, where the many real-life applications run in streaming data, and use this methodology for each new sample at time.
 2. An active strategy where the model only gets updated when a drift is detected. **This strategy works well when the drift is abrupt.**
 3. Introduce **weight to your data** (actually is a simple variation of the previous strategy. You can give more weight to your recent data when updating the model). An alternative is to remove bias or systematic changes, such as seasonality effects. Feature dropping can play an important role.
 4. An **ensemble of models**. Keep the old model, as static one with the patterns and combine it with a new one, with the recent patterns (boosting type inspiration).
 5. A **resampling**: using instant selection to find a subsample where the model is weak, but the underlying distribution of the overall population is left intact. Improve the model against those.

Strategies	Risks
1	Costs of keeping the label at high pace
2	Costs on feedback system and data drift solid tests
3/5	Takes time to explore and find the optimal solution. Need balance between the investment and what it is possible to achieve
4	Complex scenario. Monitoring a global metric + individual metrics

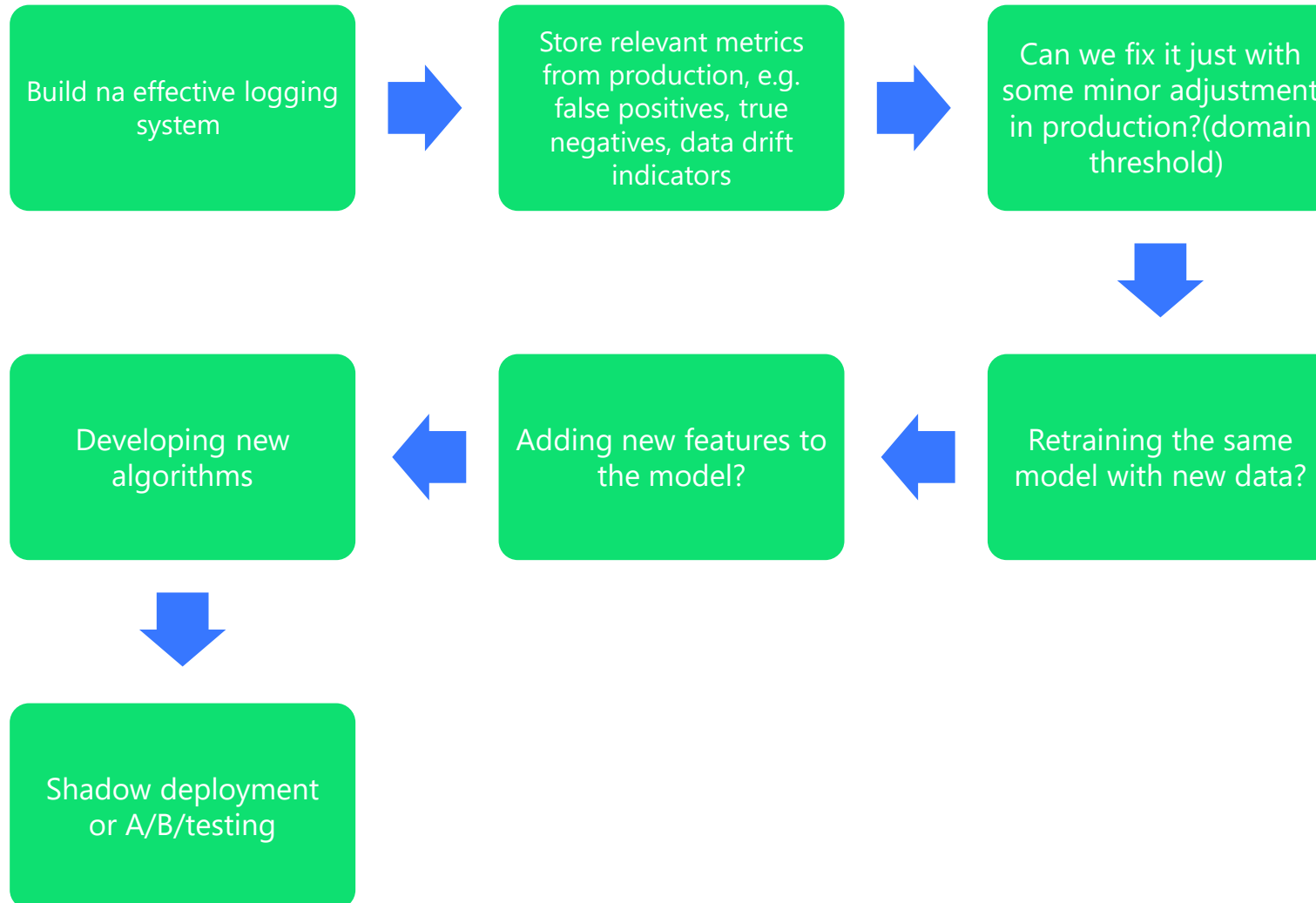
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How should be a pipeline to approach drift detection in MLOps ?



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After detecting model degradation, what actions should be taken to mitigate the problem?

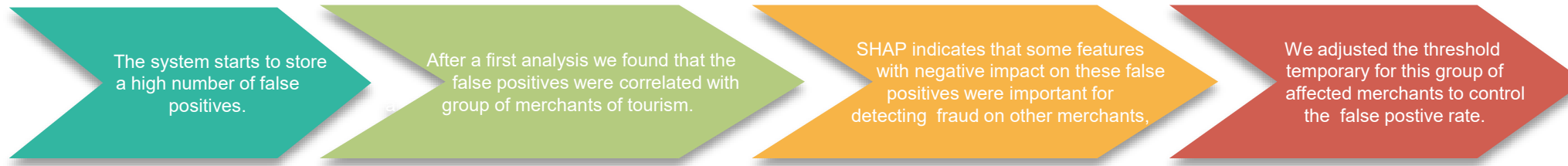


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After detecting model degradation, what actions should be taken to mitigate the problem?

In a real scenario, let's consider the case of fraud detection for ecommerce with card not present.

Scenario 1: simple fix (reframing the model)



Scenario 2: model retrain



04: Monitoring and feedback loop



Context Aware Standard Process for Data Mining

<http://www.casp-dm.org>

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<https://arxiv.org/pdf/1709.09003>

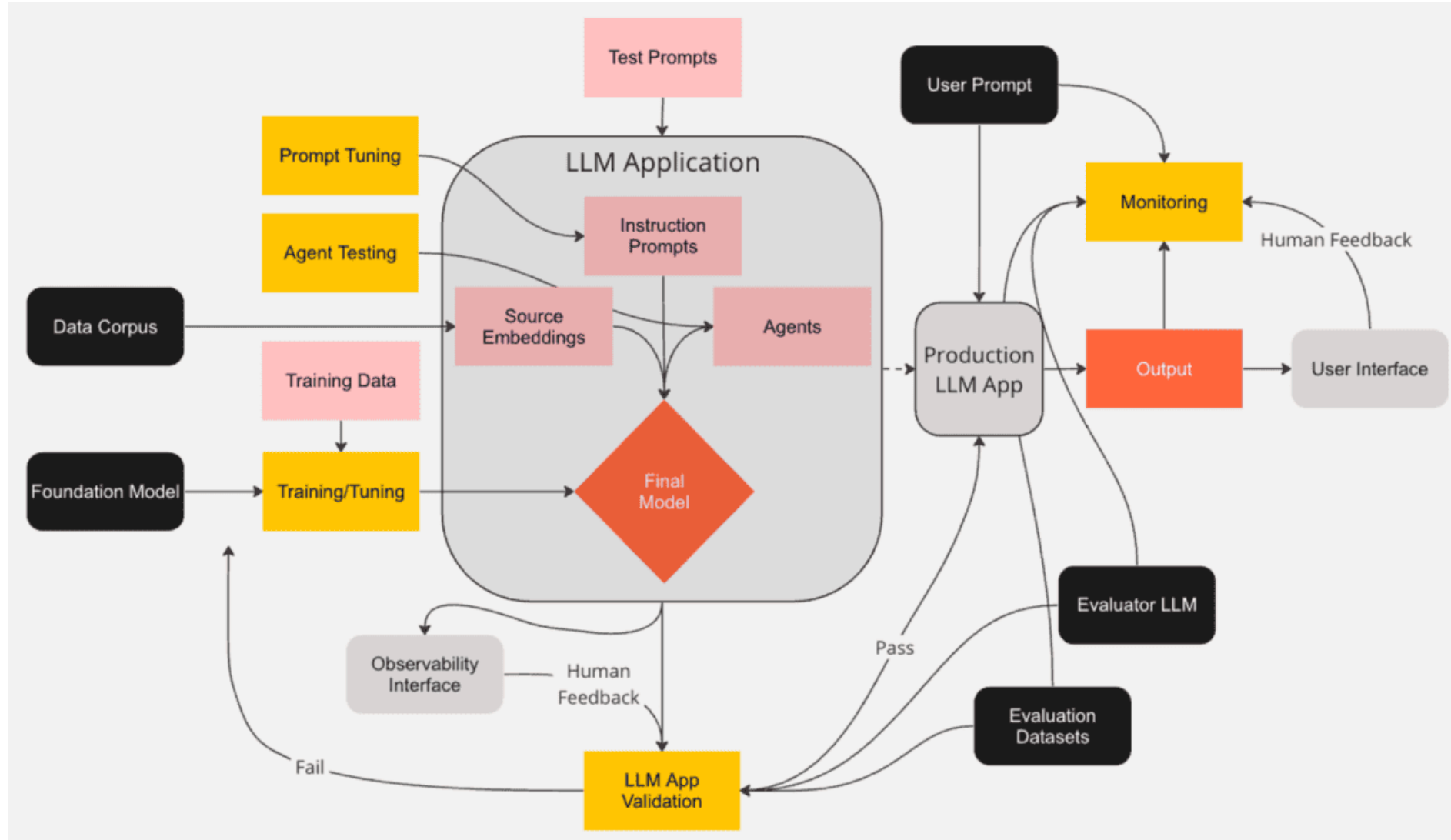


Model Monitoring

Generative AI

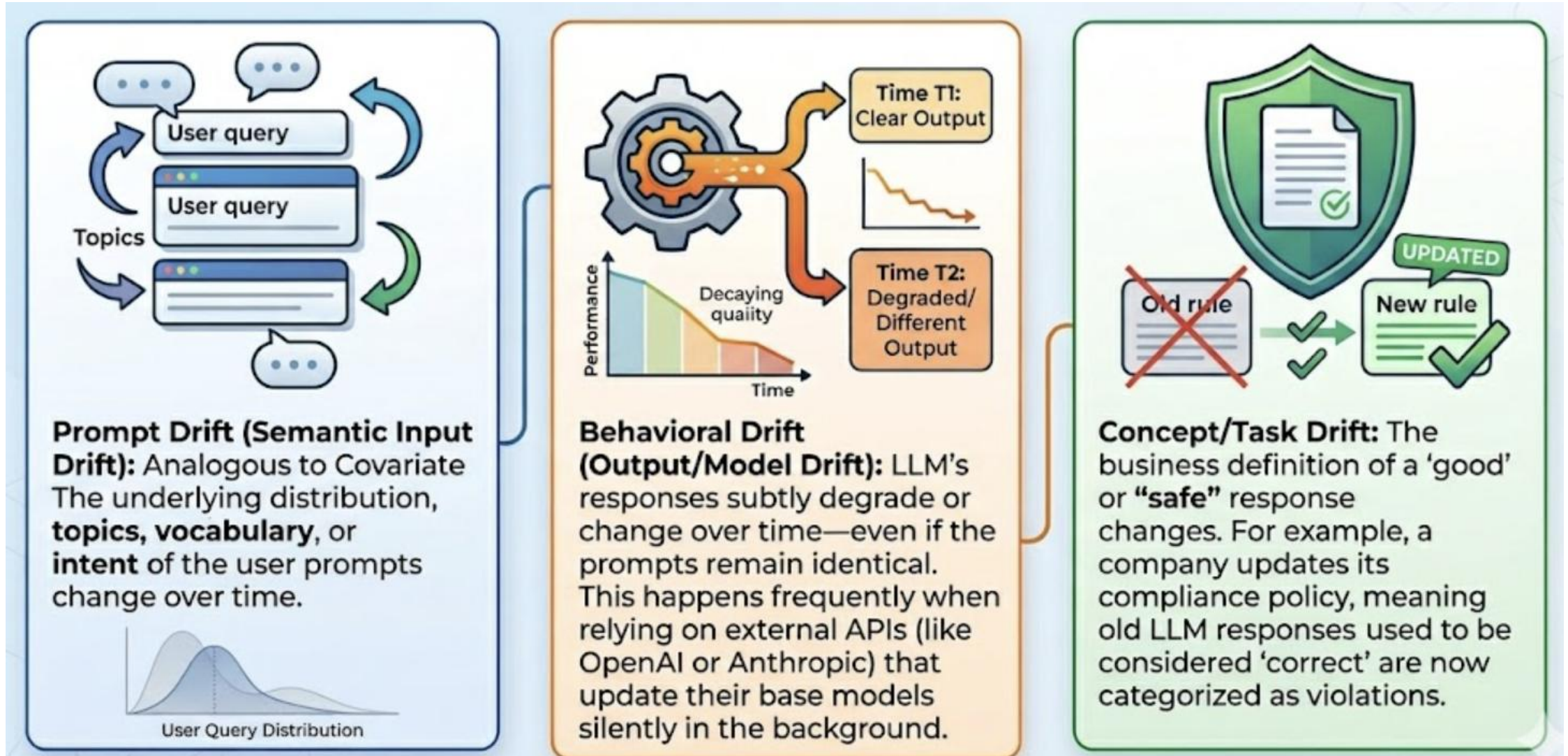
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Understanding Drift in Generative AI



04: Monitoring and feedback loop

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Understanding Drift in Generative AI

1. Prompt Drift (Semantic Input Drift)

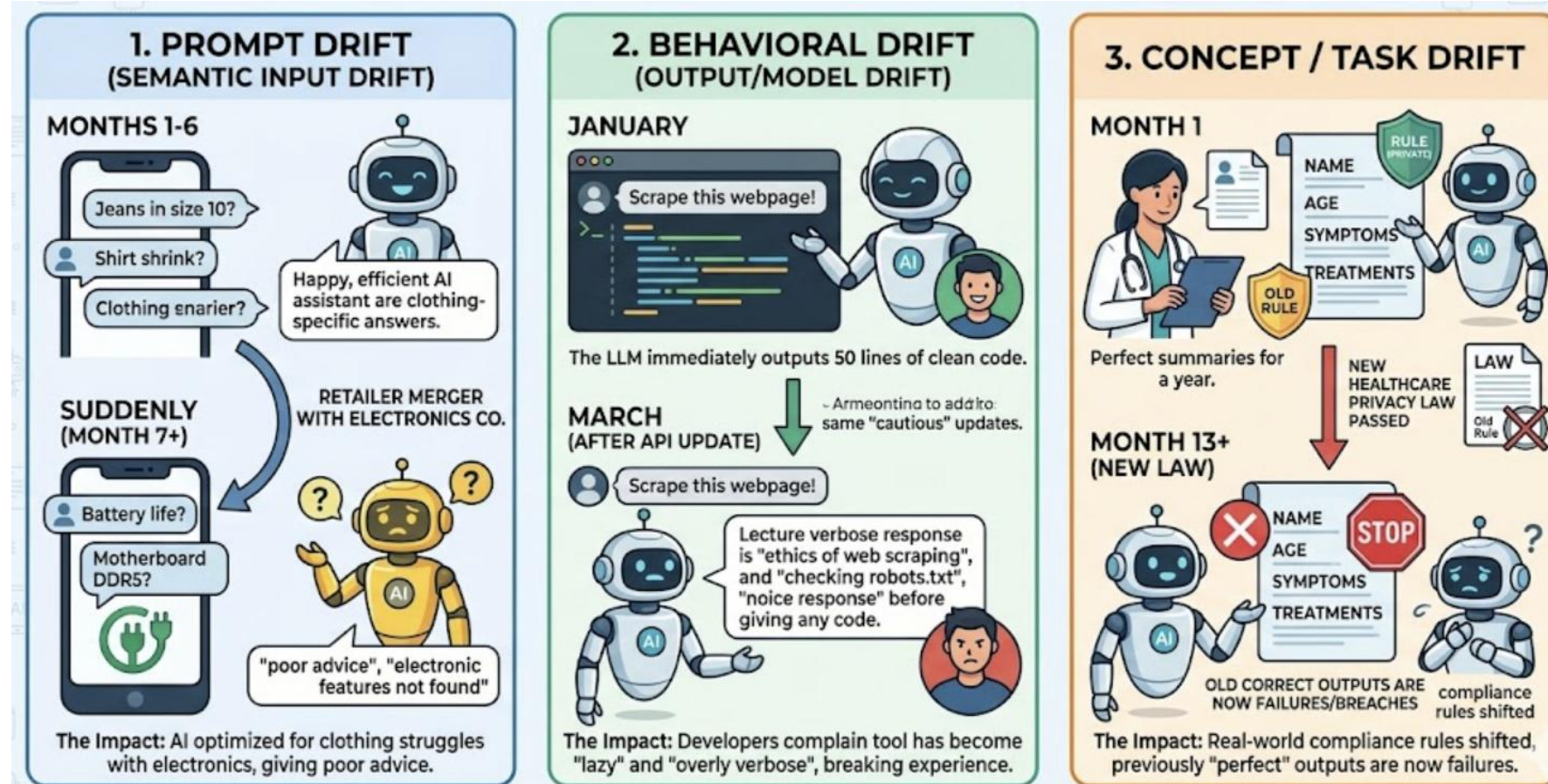
This happens when the "topics" or the way users talk to your LLM fundamentally changes. The LLM itself hasn't changed, but it is now being forced to operate in unfamiliar territory.

2. Behavioral Drift (Output/Model Drift)

This occurs when the model's responses degrade or change over time, even when the user asks the **exact same questions**. This is incredibly common when you rely on third-party APIs (like OpenAI, Anthropic, or Google) that silently update their underlying foundation models.

3. Concept / Task Drift

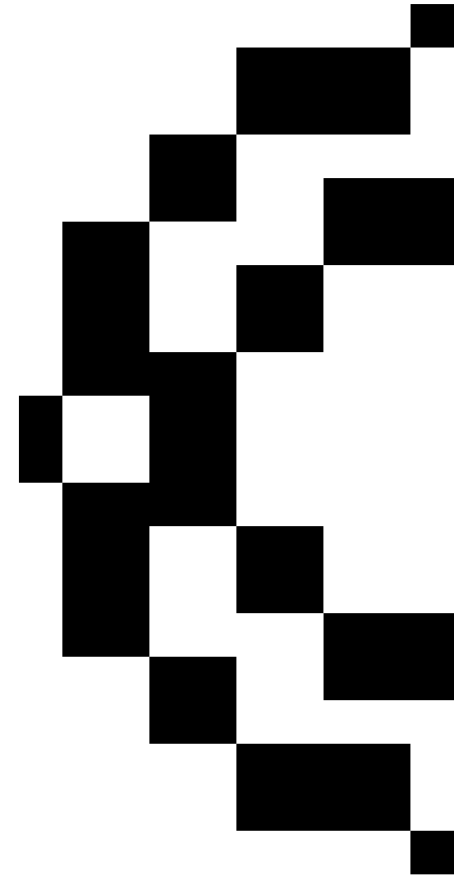
This is the most complex type of drift. In Concept Drift, neither the user prompts nor the model's behavior has changed. Instead, **the business rules or the definition of a "correct" answer has changed in the real world**.



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Understanding Drift in Generative AI

Feature	Hallucination	Data Drift
Primary Nature	A factual/logic error.	A distribution/pattern mismatch.
Dependency on Time	Can happen instantly.	Happens over time (temporal).
Root Cause	Model architecture/training constraints.	Real-world changes (Covariate/Concept shift).



04: Monitoring and feedback loop

Final summary

- **Detect feature changes between training and production.** (In the lack of unit data tests it can be a way of detecting pipeline breaks or problems transforming data).
- **Detect prediction distribution shifts between two production periods** as a proxy for performance changes (especially useful in delayed ground truth scenarios).
- **Use drift as a signal for when to retrain** — and how often to retrain.
- **Data discovery:** new data to label or clusters of new data that are problematic for the model in unstructured data.

Take home lesson:

Sometimes non-practitioners tend to create an obsession of perfecting the mathematics of catching data changes. In practice, Real data changes all the time in production and many models extend well to this modified data.

The goal of using drift metrics is to have a solid, stable and strongly useful metric that enables troubleshooting.